

# UnMask: Supplementary Material

## Conversation Transcripts (Appendix A) and VLM Session (Appendix C)

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### Appendix A: Conversation Transcripts

The P3 specification requires 5 conversation transcripts demonstrating no answer leak before the 3rd turn, plus multimodal VLM demonstration (Appendix C). All transcripts were generated live from UnMask with cold-start mastery = 0.20 (PCR `context_only` mode). Gold answers are shown for grader reference only; they were present in `InternalAnalysis` (hidden) and absent from `VisibleResponse` (shown to student). Leak status verified by both layers of the answer-leak detector.

#### Transcript 1 — Brachial Plexus Origins

**Turn 1** | PCR: `context_only` | Leak: None | Ends ‘?’: Yes

*Student:* What spinal cord levels form the brachial plexus?

*UnMask:* Great question! Let’s explore the spinal levels involved. Which specific cervical and thoracic spinal nerves contribute to the formation of the brachial plexus, and how do they relate to the vertebrae they emerge from?

[Gold held internally: “C5–T1 anterior rami”]

**Turn 2** | PCR: `context_only` | Leak: None | Ends ‘?’: Yes

*Student:* Is it only cervical nerves?

*UnMask:* You’re on the right track! Have you considered whether any thoracic spinal nerves might also contribute to the brachial plexus, in addition to the cervical nerves?

[Gold held internally: “Lower 4 cervical (C5–C8) and first thoracic (T1)”]

**Turn 3** | PCR: `prerequisite_first` | Leak: None | Ends ‘?’: Yes

*Student:* So five roots total from C5 to T1?

*UnMask:* You’re doing well in identifying the spinal roots involved! How do these spinal nerve roots organize themselves into the trunks and divisions of the brachial plexus?

[Gold confirmed: “Yes—C5, C6, C7, C8, T1”]

#### Transcript 2 — Axillary Nerve & Shoulder Dislocation

**Turn 1** | PCR: `context_only` | Leak: None | Ends ‘?’: Yes

*Student:* Which nerve is damaged when a patient can’t abduct past 15 degrees after shoulder dislocation?

*UnMask:* What is the primary muscle responsible for shoulder abduction beyond 15 degrees, and which nerve innervates this muscle?

[Gold held: “Axillary nerve”]

**Turn 2** | PCR: `context_only` | Leak: None | Ends ‘?’: Yes

*Student:* So is it the axillary nerve?

*UnMask:* You’re on the right track! What is the primary muscle responsible for shoulder abduction beyond 15 degrees, and which nerve innervates this muscle?

[Gold held: “Axillary nerve innervates deltoid, crucial for abduction beyond 15°”]

**Turn 3** | PCR: `prerequisite_first` | Leak: None | Ends ‘?’: Yes

*Student:* What else does the axillary nerve innervate besides deltoid?

*UnMask:* Besides the deltoid, which other muscle and what specific area of skin does the axillary nerve innervate?

[Gold confirmed: “Teres minor + lateral shoulder skin (regimental badge area)”]

#### Transcript 3 — Radial Nerve & Wrist Drop

**Turn 1** | PCR: `context_only` | Leak: None | Ends ‘?’: Yes

*Student:* A patient has wrist drop after a mid-shaft humerus fracture. What nerve is likely injured?

*UnMask:* What is the relationship between the location of a mid-shaft humerus fracture and the nerve that runs in that area, particularly in relation to wrist extensor function?

[Gold held: “Radial nerve”]

**Turn 2** | PCR: context\_only | Leak: None | Ends ‘?’: Yes

*Student:* Does the radial nerve innervate all wrist extensors?

*UnMask:* What is the role of the radial nerve in the forearm, and how does it relate to the muscles responsible for wrist extension?

[Gold held: “Radial nerve innervates all wrist extensor muscles”]

**Turn 3** | PCR: prerequisite\_first | Leak: None | Ends ‘?’: Yes

*Student:* What about sensation: where would the patient feel numbness?

*UnMask:* Which specific area on the dorsal side of the hand and forearm would you expect to experience numbness if the radial nerve is injured?

[Gold confirmed: “Posterior forearm and dorsal hand; first dorsal web space”]

#### Transcript 4 — Rotator Cuff Anatomy

**Turn 1** | PCR: context\_only | Leak: None | Ends ‘?’: Yes

*Student:* Name the four rotator cuff muscles.

*UnMask:* What acronym can help you remember the names of the four muscles that form the rotator cuff, and what does each letter stand for?

[Gold held: “Supraspinatus, infraspinatus, teres minor, subscapularis (SITS)”]

**Turn 2** | PCR: context\_only | Leak: None | Ends ‘?’: Yes

*Student:* I know supraspinatus, but what about the others on the posterior scapula?

*UnMask:* What are the names of the rotator cuff muscles that originate from the infraspinous fossa and the teres minor region on the posterior scapula?

[Gold held: “Infraspinatus and teres minor”]

**Turn 3** | PCR: prerequisite\_first | Leak: None | Ends ‘?’: Yes

*Student:* What movement does the infraspinatus produce?

*UnMask:* Considering the infraspinatus is located behind the glenohumeral joint, how might its position influence the type of rotation it facilitates at the shoulder?

[Gold confirmed: “External rotation of the shoulder”]

#### Transcript 5 — Median Nerve & Carpal Tunnel Syndrome

**Turn 1** | PCR: context\_only | Leak: None | Ends ‘?’: Yes

*Student:* What nerve is compressed in carpal tunnel

syndrome?

*UnMask:* Which nerve passes through the carpal tunnel along with the flexor tendons and is responsible for sensation in the thumb, index, middle, and part of the ring finger?

[Gold held: “Median nerve”]

**Turn 2** | PCR: context\_only | Leak: None | Ends ‘?’: Yes

*Student:* What symptoms would the patient have?

*UnMask:* When the median nerve is compressed at the wrist, what changes might you expect in both sensation and muscle function in the hand?

[Gold held: “Numbness/tingling digits 1–3.5; thenar weakness”]

**Turn 3** | PCR: prerequisite\_first | Leak: None | Ends ‘?’: Yes

*Student:* Would the patient have weakness gripping or just sensory loss?

*UnMask:* What role do the thenar muscles, innervated by the median nerve, play in gripping, and how might their dysfunction affect hand strength?

[Gold confirmed: “Both—thenar weakness impairs pinch/grip; sensory loss in lateral digits”]

**Transcript summary:** 15 turns, 5 transcripts, 0 confirmed leaks, 0 soft flags, 15/15 ends with “?”. Gold answer present only in InternalAnalysis (stripped before rendering); absent from retrieved Qdrant chunks due to PCR must\_not filter.

## Appendix B: Onboarding, Visual Aid, and Session Summary

### B.1 Personalized Onboarding and Diagnostic Reordering

The welcome prompt asks a single conversational question; the student’s reply is parsed for study\_focus and learning\_mode before the LangGraph graph is invoked.

#### UnMask welcome:

“Hey! Welcome to UnMask. I’m your NBCOT anatomy study companion. I use the Socratic method to help you build clinical reasoning, not just recall facts. What are you finding toughest right now, and do you prefer learning with diagrams or written Q&A? (Just reply naturally, I’ll calibrate from there!)”

*Student:* “I keep getting confused by the brachial plexus, especially which levels form it. I think I learn better with diagrams.”

**System action** (Orchestrator, before graph.invoke):

```
study_focus = "brachial plexus"
learning_mode = "visual"
diag_order = [0, 2, 3, 1] # BP questions first
visual_hint_threshold = 1 # diagram on 1st wrong
```

**Diagnostic Q1 (reordered)** | Phase: rapport  
*UnMask*: “Quick diagnostic – 4 questions to calibrate where we start: **Q1 of 4**: Which spinal cord levels give rise to the brachial plexus?”  
*Student*: “C5 to T1” → **Correct**.  
mastery[brachial\_plexus.origin] = 0.50

## B.2 Visual Aid Trigger (Gray’s Anatomy Plate)

Visual mode threshold is 1. Below: first incorrect answer in Tutoring phase.

**Turn 1** | PCR: context\_only | Phase: tutoring

*UnMask*: “How do the C5–T1 roots combine to form the three trunks of the brachial plexus?”

*Student*: “I’m not sure.” → **Incorrect**.

```
consecutive_incorrect = 1 # >= threshold (1)
```

**System action** (Gray809, public domain, 108 KB PNG):

```
cl.Image(path="public/anatomy/brachial_plexus.png",
display="inline") # appended to reply
```

**UnMask reply with inline plate:**

*“That’s a tricky one to visualize! Here is the brachial plexus diagram for reference. [Gray809 image renders inline.] Look at where C5–C6 converge: what structure do they form at that junction?”*

## B.3 Session Wrapup Summary (Structured Output)

At wrapup ( $t \geq 840$ s), GPT-4o structured output produces a SessionSummary:

```
overall_assessment:
  "Solid C5-T1 recall and correct radial wrist drop.
  Trunk formation and teres minor need more work."

topic_reports:
- concept: brachial_plexus.trunks
  mastery_score: 0.25
  status: needs_review
  honest_feedback: "Struggled turns 1+4; review
  how C5-C6 form the upper trunk."
- concept: brachial_plexus.origin
  mastery_score: 0.50
  status: progressing
  honest_feedback: "Root levels correct; reinforce
  anterior rami distinction."
- concept: peripheral_nerves.radial
  mastery_score: 0.72
  status: mastered
  honest_feedback: "Wrist drop identified correctly
  with good clinical reasoning."

mistake_highlights:
- "Confused upper trunk (C5+C6) with C5 alone."
- "Missed teres minor (2nd axillary target)."
```

```
study_recommendations:
- "Draw RTDC (Roots-Trunks-Divisions-Cords)."
- "Practice Phalen’s and Tinel’s sign sequences."

closing_reflection:
  "Patient has deltoid weakness and lateral arm
  numbness post-dislocation. Which nerve and cord
  would you evaluate first?"
```

The mistake\_highlights field is populated from mistake\_log entries whose misconception strings were captured at the time of each incorrect answer (hidden InternalAnalysis.student\_misconception). The closing\_reflection ends with “?” – the session ends on a Socratic note, not a declarative summary.

## Appendix C: Multimodal VLM Session (Task 4)

**Context:** Student uploads a photo of a labeled anatomical diagram of the brachial plexus during the tutoring phase (consecutive\_incorrect = 1, PCR mode: context\_only).

**System action** (analyze\_uploaded\_image, GPT-4o Vision):

```
Image received: student_upload_brachial.jpg
VLM internal: "Identified structure: Brachial plexus
diagram
  showing C5-T1 roots, three trunks (upper/middle/
  lower),
  and lateral/medial/posterior cords."
VLM output (Socratic, structure unnamed):
  "I can see a nerve network diagram here --- look
  at the
  connections between the spinal levels at the top
  and
  the groupings below them. What do you notice
  about
  how the upper levels combine compared to the
  lower
  ones, and what might that tell you about how
  injury
  at different levels would produce different
  symptoms?"
```

**Turn 1** | Source: student image upload | VLM: GPT-4o Vision | Leak: None

*Student*: [uploads anatomical image]

*UnMask (VLM)*: “I can see a nerve network diagram here — look at the connections between the spinal levels at the top and the groupings below them. What do you notice about how the upper levels combine compared to the lower ones, and what might that tell you about how injury at different levels would produce different symptoms?”

[Structure identified internally: brachial plexus roots/trunks — not named in response]

**Turn 2** | Phase: tutoring | PCR: context\_only | Leak: None

*Student*: “The upper levels C5 and C6 seem to merge together.”

*UnMask*: “Exactly right — they do converge. Now,

if that convergence point were injured, which specific movements of the arm would you expect to be affected, and why?"

*[Gold held: "C5+C6 form upper trunk; injury causes Erb's palsy — shoulder abduction/external rotation lost"]*

**Turn 3** | PCR: prerequisite\_first | Leak: None

*Student:* "Shoulder movement would be lost?"

*UnMask:* "You're on the right track! Specifically, which shoulder movements depend on C5 and C6, and what clinical presentation would you expect in a newborn with a difficult delivery?"

*[Gold confirmed: "Waiter's tip posture (Erb's palsy) — abduction and external rotation lost"]*

**VLM performance note:** The structure was identified without being named in the Socratic question (0 leaks). The VLM response maintained Socratic form (ends with "?") and guided the student toward discovering the convergence pattern independently. This completes Task 4 (Multimodal Diagram Tutoring) from the project specification.